

NMAP “scanme.nmap.org” REPORT

Target of the Report

This report focuses on evaluating the security risks of open ports found in “scanme.nmap.org”. The main goal is to analyze which services are exposed to the network, assess the vulnerabilities linked to the open ports, and how they could be exploited by the attackers. By identifying the potentially vulnerable ports this aims to improve the system’s overall security.

Testing Methodology

- Network Scanning using Nmap
 - Use of different scanning types
 - Regular scan.
 - Quick scan
 - UDP scan
 - Intense scan
 - SYN scan
 - Full TCP port scan *-p-*
 - Aggressive scan *-A*
- Manual Review/Research
 - Searching articles for Information.
 - Port usage
 - Common Vulnerabilities and Exposures (CVEs) / Vulnerabilities
 - (cve.mitre.org)
 - Security Impacts
 - Mitigating Recommendations
- Documentation

- Google Docs

Ports	Why the Port is Used	vulnerabilities	Type of Scan	Impact	Recommendations
Port: 22, SSH - Secure Shell, Open	This is used to provide secure, encrypted remote access for file transfers, system administration, and automation. For safe communication over networks.	• Brute-force attacks / CVE-2015-5600 • Privilege escalation / CVE-2021-41617 • System compromise • Information leak / CVE-2016-0777	Quick Scan, “nmap -T4 -F scanme.nmap.org”	• significantly increase cyberattacks if not properly secured • Leaked SSH keys (unauthorized access attempts).	1. Disable password authentication. Use SSH key Authentication. 2. Disable root login. 3. Use firewall to restrict access 4. Keep server updated 5. Use intrusion tools.

Port: 80, HTTP - HyperText Transfer Protocol, Open	HTTP is mostly used for websites that don't need encryption or don't have confidential data.	• Server Misconfigurations • SQL Injection • XSS Cross-site scripting	Quick Scan, "nmap -T4 -F scanme.nmap.org"	• Unencrypted data, vulnerable to interception and data stealing . • No data integrity • Data manipulation	1. Switch to HTTPS port 443. 2. Regularly update and patch web servers. 3. Enable Web Application Firewall.
Port: 123, NTP - Network Time Protocol, Open	Used to synchronize clocks between computer systems over a network.	• NTP amplification attack (DDoS) / CVE-2013-5211 • Authenticated access • Information leaks	UDP Scan, "nmap -sU scanme.nmap.org"	• Can be exploited in DDoS amplification attacks. • Information disclosure • Time tampering (can lead to authentication or log integrity)	1. Restrict access to port 123 to trusted IPs or internal networks only. 2. Keep NTP software updated to mitigate vulnerabilities. 3. Use authentic

				y issues).	ated NTP if security is critical.
Port: 9929, Nping-Echo, Open	Used by network administrators for ping-like tests, but more flexible than ICMP echo requests. Helps testers understand how a network behaves under different packet conditions, useful for vulnerability and security assessments	● Potential info gathering	Regular Scan “nmap scanme.nmap. org”	● If exposed, it may indicate that Nping is running on a system, and it could be used by attackers to map network layouts, analyze traffic, and detect vulnerabilities. ● Attackers can use this port to send custom	1. Block Port 9929. 2. Disable Nping Echo Service. 3. Monitor network traffic for suspicious activity.

				packets, which may cause denial of service or disrupt network services.	
Port: 31337, tcpwrapped, Open	A remote access Trojan. Allows attackers to remotely control a Windows system. Nowadays it is commonly flagged in security audits due to its link with malware and hacker tools.	• Known backdoor / CVE-1999-0504 • Associated with malwares.	Regular Scan “nmap scanme.nmap.org”	• It is open and running but is actively blocking access from unauthorized IPs. • Could be a sign of a backdoor or malicious service. • Port 31337	1. Block port 31337 2. Monitor for any malicious activity 3. Scan for any malware 4. Patch and secure systems

				is strongl y associat ed with malwar e. Even when tcpwrap ped, its presenc e should be reviewe d.	
--	--	--	--	--	--

Summary of the Report

The scan of “scanme.nmap.org” revealed several open ports that could expose the system to risks. Each open port was analyzed to see what service it runs, what known security issues it may have, and how serious the impact could be if someone tried to exploit it.

The scan found that some ports like SSH(22) and HTTP(80), are commonly used but it can be risky if not properly secured. Other ports, like the NTP(123) and Nping Echo(9929), are less common but still pose threats if left unmonitored or open to the public. One port (31337) which was “tcpwrapped”, meaning that it may be blocked but is still worth taking a look at, as it’s often linked with suspicious activity.

Based on the findings, the report gives steps or recommendations to reduce risk. Overall, the scan showed that even commonly used ports can be dangerous without proper configuration. This report helps identify the risks and gives recommendations for mitigation that are needed to better secure the system.

Evidences

Normal / Regular Scan

```
(kali㉿kali)-[~]  
$ nmap scanme.nmap.org  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-28 09:03 EDT  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.27s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 995 closed tcp ports (reset)  
PORT      STATE      SERVICE  
22/tcp    open       ssh  
25/tcp    filtered   smtp  
80/tcp    open       http  
9929/tcp  open       nping-echo  
31337/tcp open       Elite  
  
Nmap done: 1 IP address (1 host up) scanned in 9.86 seconds
```

Quick Scan

```
(kali㉿kali)-[~]  
$ nmap -T4 -F scanme.nmap.org  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-28 09:05 EDT  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.27s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 97 closed tcp ports (reset)  
PORT      STATE      SERVICE  
22/tcp    open       ssh  
25/tcp    filtered   smtp  
80/tcp    open       http  
  
Nmap done: 1 IP address (1 host up) scanned in 2.39 seconds  
  
(kali㉿kali)-[~]  
$
```

UDP Scan

```
(kali㉿kali)-[~]  
$ nmap -sU scanme.nmap.org  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-24 05:03 EDT  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.023s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 999 open|filtered udp ports (no-response)  
PORT      STATE      SERVICE  
123/udp   open       ntp  
  
Nmap done: 1 IP address (1 host up) scanned in 15.24 seconds  
  
(kali㉿kali)-[~]  
$
```


Intense Scan

```
Not shown: 995 closed tcp ports (reset)
PORT      STATE      SERVICE      VERSION
22/tcp    open       ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
| 1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
| 2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
| 256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_ 256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
25/tcp    filtered  smtp
80/tcp    open       http         Apache httpd 2.4.7 ((Ubuntu))
|_ http-title: Go ahead and ScanMe!
|_ http-favicon: Nmap Project
|_ http-methods:
|_ Supported Methods: GET HEAD POST OPTIONS
|_ http-server-header: Apache/2.4.7 (Ubuntu)
9929/tcp  open       nping-echo   Nping echo
31337/tcp open       tcpwrapped
Aggressive OS guesses: Linux 4.19 - 5.15 (98%), Linux 4.15 (95%), IPFire 2.27 (Linux 5.15 - 6.1) (94%), Linux 5.4 (94%), Linux 91%), Linux 3.11 - 4.9 (91%), Linux 3.2 - 3.8 (91%), Linux 2.6.32 (91%), Android TV OS 11 (Linux 4.19) (90%)
No exact OS matches for host (test conditions non-ideal).
Uptime guess: 33.403 days (since Tue Jun 24 23:50:19 2025)
Network Distance: 10 hops
TCP Sequence Prediction: Difficulty=258 (Good luck!)
IP ID Sequence Generation: All zeros
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE (using port 587/tcp)
HOP RTT      ADDRESS
1   8.39 ms  eLifeConnect (192.168.1.1)
2   13.26 ms bba-94-59-84-1.alshamil.net.ae (94.59.84.1)
3   13.26 ms  10.246.255.29
4   14.45 ms  194.170.186.168
5   272.76 ms 195.229.0.149
6   ... 9
10  276.72 ms scanme.nmap.org (45.33.32.156)
```

SYN Scan all TCP ports

```
(kali㉿kali)-[~]
$ nmap -sS -p- scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-28 09:10 EDT
Stats: 0:00:03 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 0.35% done
Stats: 0:00:07 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 0.96% done
Stats: 0:02:45 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 27.02% done; ETC: 09:20 (0:07:26 remaining)
Verbosity Increased to 1.
SYN Stealth Scan Timing: About 94.26% done; ETC: 09:20 (0:00:36 remaining)
Completed SYN Stealth Scan at 09:21, 648.50s elapsed (65535 total ports)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.27s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 65530 closed tcp ports (reset)
PORT      STATE      SERVICE
22/tcp    open       ssh
25/tcp    filtered  smtp
80/tcp    open       http
9929/tcp  open       nping-echo
31337/tcp open       Elite

Read data files from: /usr/share/nmap
Nmap done: 1 IP address (1 host up) scanned in 648.92 seconds
Raw packets sent: 67439 (2.967MB) | Rcvd: 68139 (2.750MB)
```

Sources

<https://www.cbtnuggets.com/common-ports/what-is-port-22>
<https://www.netcomlearning.com/blog/what-is-port-22>
<https://www.cbtnuggets.com/common-ports/what-is-port-80>
<https://web.synametrics.com/is-port-80-dangerous-to-open.htm>
<https://www.speedguide.net/port.php?port=123>
<https://www.cbtnuggets.com/common-ports/what-is-port-123>
<https://www.speedguide.net/port.php?port=9929>
<https://www.cbtnuggets.com/common-ports/what-is-port-31337>
<https://www.sshhandbook.com/security-best-practices/>
<https://medium.com/%40redswitches/secure-ssh-connections-7-best-practices-you-should-apply-right-now-e97f8da72bc0>
<https://www.cve.org/>
<https://cve.mitre.org/cgi-bin/cvekey.cgi?keyword=openssh>
<https://superuser.com/questions/1712116/nmap-shows-excessive-number-of-open-ports?>

